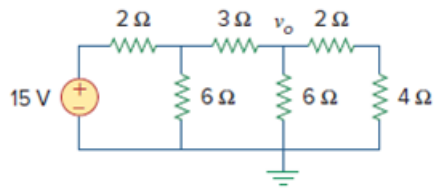


## Problem

For the circuit in Fig. 4.73, assume  $v_o = 1$  V, and use linearity to find the actual value of  $v_o$ .

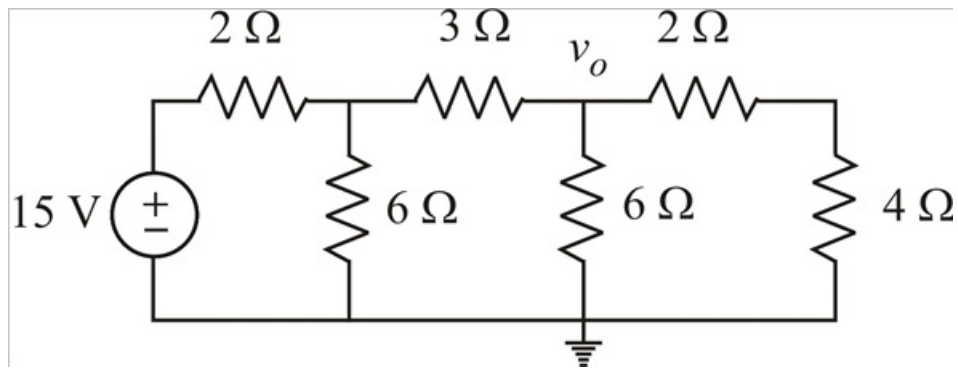
**Figure. 4.73:**



## Step-by-step solution

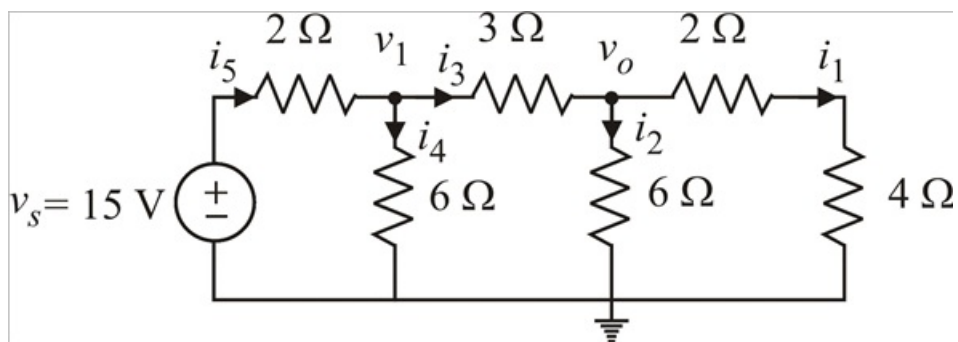
## Step 1 of 6

The given circuit is,



## Step 2 of 6

Using linearity property,



## Step 3 of 6

Assuming  $v_0 = 1\text{ V}$ ,

$$i_1 = \frac{v_0}{2+4}$$

$$i_1 = \frac{1}{6}\text{ A}$$

And

$$i_2 = \frac{v_0}{6}$$

$$i_2 = \frac{1}{6}\text{ A}$$

#### Step 4 of 6

Applying KCL at node  $v_0$  gives,

$$i_3 = i_1 + i_2$$

$$i_3 = \frac{1}{6} + \frac{1}{6}$$

$$i_3 = \frac{1}{3}\text{ A}$$

$$\therefore v_1 = v_0 + 3i_3$$

$$v_1 = 1 + 3\left(\frac{1}{3}\right)$$

$$v_1 = 2\text{ V}$$

#### Step 5 of 6

$$i_4 = \frac{v_1}{6}$$

$$i_4 = \frac{2}{6}$$

$$i_4 = \frac{1}{3}\text{ A}$$

Applying KCL at node  $v_1$  gives,

$$i_5 = i_3 + i_4$$

$$i_5 = \frac{1}{3} + \frac{1}{3}$$

$$i_5 = \frac{2}{3}\text{ A}$$

#### Step 6 of 6

$$\therefore v_s = v_1 + 2i_s$$

$$v_s = 2 + 2\left(\frac{2}{3}\right)$$

$$v_s = \frac{10}{3} \text{ V}$$

This shows that assuming  $v_0 = 1 \text{ V}$  gives  $v_s = \frac{10}{3} \text{ V}$ .

The actual voltage source of  $v_s = 15 \text{ V}$  will give,  $v_0 = \frac{3}{10} \times 15$

$$\boxed{v_0 = 4.5 \text{ V}}$$